

University Institute of Technology - RGPV, Bhopal

Department of Information Technology



GD Seminar Presentation

Topic: Cybersecurity in 6G Networks

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Understanding 6G Technology

6G promises speeds 100x faster than 5G, delivering terabit-per-second data rates with ultra-low latency under 1 millisecond. This transformative leap will power AI-native networks that fundamentally reshape how we connect.

Game-Changing Applications

Fully autonomous vehicles with instant response

Immersive holographic communications

Smart cities with real-time intelligence

Massive IoT ecosystems connecting trillions of devices





**Why
Cybersecurity
Matters More
Than Ever in 6G**



Massive Scale

6G will connect billions more devices than ever before, creating an exponentially larger attack surface for potential threats.



Sophisticated Attacks

As networks evolve, cybercriminals develop more advanced techniques to exploit vulnerabilities and breach systems.



Data Protection

With more sensitive personal and business data flowing through networks, protecting privacy becomes increasingly complex and critical.

or create backdoors for attackers.

Critical Security Challenges Ahead

- **Expanded Attack Surface**
More connected devices mean exponentially more entry points for hackers to exploit and compromise network security.
- **Data Privacy Concerns**
Protecting sensitive personal information becomes harder as data flows across more platforms and devices simultaneously.
- **Device Authentication**
Identifying and preventing fake or cloned devices from accessing the network poses a growing challenge.
- **AI-Powered Threats**
Cybercriminals leveraging artificial intelligence to create more sophisticated, adaptive, and harder-to-detect attacks.
- **IoT Vulnerabilities**
Many Internet of Things devices lack robust security features, creating weak points throughout the network.

Common Cyber Threats in 6G Networks



→ Data Theft

Unauthorized access to steal valuable personal or corporate information from network systems.

→ Identity Theft

Criminals impersonating users to gain access to accounts, services, and sensitive resources.

→ System Hacking

Breaking into networks and devices to control systems or extract confidential data.

→ DDoS Attacks

Overwhelming networks with traffic to shut down services and disrupt operations.

→ Malware & Viruses

Malicious software designed to damage systems, steal data, or create backdoors for attackers.

Advanced Security Technologies for 6G



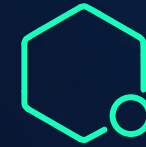
Strong Encryption

Advanced cryptographic methods that scramble data, making it unreadable to unauthorized users even if intercepted during transmission.



Secure Authentication

Multi-factor verification and biometric systems ensuring only authorized users and devices can access the network.



Blockchain Trust

Distributed ledger technology creating tamper-proof records and establishing trust without central authorities.



Continuous Monitoring

Real-time surveillance systems that track network activity 24/7 to detect and respond to threats instantly.



AI Threat Detection

Machine learning algorithms that identify patterns, predict attacks, and automatically respond to emerging security threats.

The Dual Role of AI in 6G Security

✓ AI as a Defender

Artificial intelligence can detect cyber threats in real-time by analyzing massive amounts of network data. It identifies unusual patterns and predicts potential attacks before they happen, enabling proactive defense measures that keep systems secure.

AI as Defender



AI as Weapon



⚠ AI as a Weapon

However, cybercriminals also leverage AI to create smarter, more adaptive attacks. AI-powered malware can learn and evolve to bypass security measures, making the cybersecurity landscape an ongoing arms race between defenders and attackers.

How Users Can Protect Themselves

01

Use Strong Passwords

Create unique, complex passwords with mixed characters and enable two-factor authentication wherever possible.

03

Avoid Unknown Apps

Only download applications from trusted sources and review permissions before installation.

05

Stay Alert Online

Never click suspicious links or open unexpected attachments that could contain malware.

02

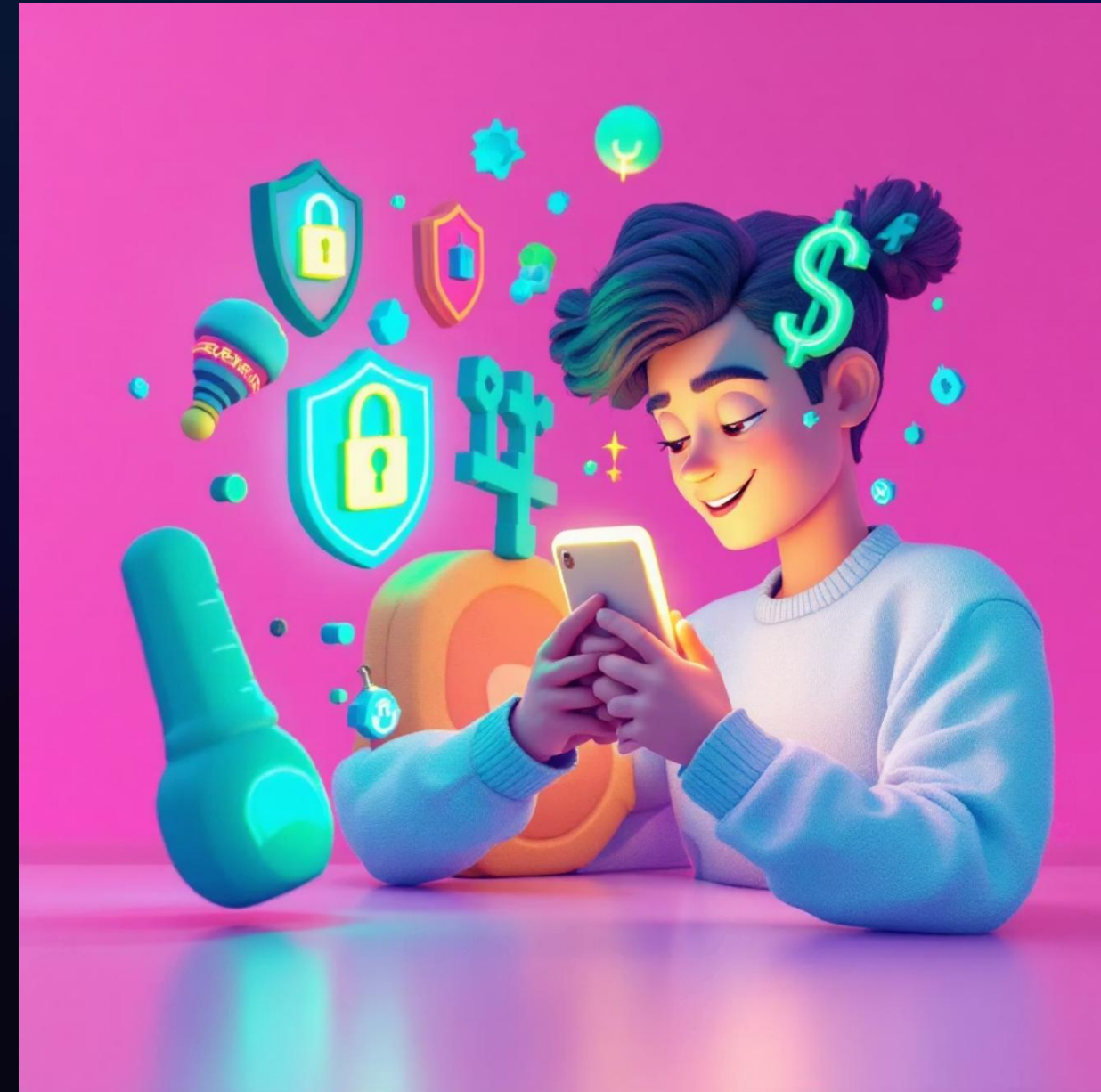
Keep Devices Updated

Regularly install software and security updates to patch vulnerabilities and protect against new threats.

04

Check Privacy Settings

Review and adjust privacy controls on all devices and accounts to limit data sharing.



Conclusion

Revolutionary Technology

6G will bring transformative capabilities that reshape how we communicate, work, and live in an interconnected world.

Security is Essential

Strong cybersecurity measures are not optional—they're fundamental to realizing 6G's full potential safely.

Protecting What Matters

Safe networks safeguard users, data, privacy, and the critical infrastructure our society depends on daily.

We must prepare today for the cyber threats of tomorrow. The future of 6G depends on our collective commitment to building secure, trustworthy networks that serve everyone safely.

Thankyou Everyone
For your attention

